Energy Density

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ENERGY DENSITY IN THE ELECTROSTATIC FIELD

- Bringing a positive charge from infinity into the field of another positive charge requires work, the work being done by the external source moving the charge.
- Energy must be conserved, and the energy expended in bringing this charge into position now represents potential energy.
- In order to find the potential energy present in a system of charges, we must find the work done by an external source in positioning the charges.

Work to position $Q_2 = Q_2 V_{2,1}$

Work to position $Q_3 = Q_3 V_{3,1} + Q_3 V_{3,2}$

Work to position $Q_4 = Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3}$

Total positioning work = potential energy of field

$$= W_E = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2} + Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3} + \cdots$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

Total positioning work = potential energy of field

$$= W_E = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2} + Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3} + \cdots$$

$$Q_3 V_{3,1} = Q_3 \frac{Q_1}{4\pi\epsilon_0 R_{13}} = Q_1 \frac{Q_3}{4\pi\epsilon_0 R_{31}}$$

$$W_E = Q_1 V_{1,2} + Q_1 V_{1,3} + Q_2 V_{2,3} + Q_1 V_{1,4} + Q_2 V_{2,4} + Q_3 V_{3,4} + \cdots$$

$$\begin{aligned} 2W_E = {} & Q_1(V_{1,2} + V_{1,3} + V_{1,4} + \cdots) \\ & + Q_2(V_{2,1} + V_{2,3} + V_{2,4} + \cdots) \\ & + Q_3(V_{3,1} + V_{3,2} + V_{3,4} + \cdots) \\ & + \cdots \end{aligned}$$

$$W_E = \frac{1}{2}(Q_1 V_1 + Q_2 V_2 + Q_3 V_3 + \cdots) = \frac{1}{2}\sum_{m=1}^{m=N} Q_m V_m$$

$$W_E = \frac{1}{2}\int_{\text{vol}} \rho_v V \, dv$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

$$W_E = \frac{1}{2}\int_{vol} \rho_v V dv = \frac{1}{2}\int_{vol} (\nabla \cdot \mathbf{D}) V \, dv$$

$$= \frac{1}{2}\int_{vol} [\nabla \cdot (V\mathbf{D}) - \mathbf{D} \cdot (\nabla V)] dv$$

$$W_E = \frac{1}{2}\oint_S (V\mathbf{D}) \cdot d\mathbf{S} - \frac{1}{2}\int_{vol} \mathbf{D} \cdot (\nabla V) \, dv$$

(Divergence theorem)

$$W_E = \frac{1}{2}\int_{vol} \mathbf{D} \cdot \mathbf{E} \, dv = \frac{1}{2}\int_{vol} \epsilon_0 E^2 \, dv$$

(First term is zero in above expression)


Thank
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